

CANONICAL DYNAMICAL GEOMETRY SYSTEM
VERSION 4 — INFORMATION-GEOMETRIC FLOW
FRactal • Contact • Geometric Flow • Fisher

ABSTRACT. We define a single, locked ordinary differential equation (the *canonical ODE*) that drives a state $X \in \mathbb{R}^n$ towards a target set encoded by a feature map $S : \mathbb{R}^n \rightarrow \mathbb{R}^k$ and a quadratic energy $E = S^T G S$, while simultaneously executing a domain-specific nominal field F_{base} . Geometry enters the dynamics *only* through the Euclidean gradient $g_X = 2J^T G S$; all projections are Euclidean; the pulled-back metric $J^T G J$ is reserved for curvature analysis. Under standard regularity (Assumptions A1–A4), Part I proves (i) local existence and uniqueness, (ii) Lyapunov descent, (iii) asymptotic stability of $\{E = 0\}$, (iv) exponential convergence with rate $\rho = 4\sigma^2 \mu_G^2 / M_G \cdot (1 - \gamma_{\max})$ under a uniform Jacobian rank condition, and (v) ultimate boundedness when $F_{\text{base}} \neq 0$, with explicit admissibility threshold and a local-in-energy capture window. Part II extends the framework to fractal targets (multi-scale mollified distance with Hausdorff-dimension weighting), contact manifolds (Reeb-aware feature map with horizontal-descent property), and time-varying metrics evolved by Ricci-type geometric flows (strict Lyapunov monotonicity for constant target metrics and practical state–metric bounds in the state-dependent case). Part III specialises the time-varying metric to feature-space Fisher information and its pullback to state space, yielding adaptive preconditioning, a chart-covariant auxiliary natural-gradient flow, and a statistical Cramér–Rao floor on the descent rate. Throughout, the canonical ODE, the gradient formula, and the Euclidean projection are unchanged; all domain content is confined to the triple (S, G, F_{base}) .

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LOCKED

Execution paradigm (LOCKED): Euclidean execution with geometric encoding. Geometry enters the dynamics *only* through the state-space gradient g_X . All projections are Euclidean. The pulled-back metric $G_{\text{pull}} = J^T G J$ is used *only* for curvature computation.

1 Introduction

Many engineering and scientific tasks reduce to the same structural problem: drive a state X along a nominal trajectory generated by F_{base} while simultaneously enforcing a geometric constraint encoded by an energy $E(X)$. Examples include statistical-arbitrage portfolios (spread $\rightarrow 0$), spectral PDE controllers (modal error $\rightarrow 0$), first-order optimisers ($\nabla f \rightarrow 0$) and task-space robotics (end-effector error $\rightarrow 0$).

Classical solutions diverge: Riemannian gradient flows replace the state-space metric and obscure execution; control-Lyapunov designs often reinvent the projection at each domain; port-Hamiltonian formulations require an energy-shaping co-design. The cost is a proliferation of incompatible formalisms.

Contribution. We commit to a single, frozen ODE (Section 4) and a single gradient (Section 5.3) that respects three inviolable rules (Section 6) and prove its properties *once*. Domain content is confined to the triple (S, G, F_{base}) (Section 11), so each application becomes an instantiation rather than a redesign.

Organisation. Section 2 records the historical origin of the kernel in the GravityEngine + BSDT pipeline and fixes the dictionary that maps every BSDT object onto a canonical symbol. Section 3 fixes assumptions and well-posedness. Sections 4–6 state the locked system. Sections 7–7 develop structural results, the curvature manifold, and the numerical implementation. Section 9 proves the main stability theorems. Section 10 instantiates the framework on four domains. Part II (Sections D–F) extends the kernel by fractal, contact, and geometric-flow constructions. Part III (Section H onwards) specialises Part II to the Fisher information metric and introduces the auxiliary natural-gradient flow.

Related work. Lyapunov-based control and CLF theory [1, 2]; geometric control [3]; gradient flows in metric spaces [4]; neural ODEs [5]; port-Hamiltonian systems [6]; adaptive control [7]. Our framework is closest to Sontag’s universal formula but differs in (a) using a fixed Euclidean projection and (b) routing all geometry through a single feature map and metric.

2 Origin: from BSDT to the canonical kernel

The canonical system was not invented in the abstract. It was extracted from a concrete pipeline — the *GravityEngine* systemic-risk simulator and its *Blind-Spot Detection Theory* (BSDT) calibration layer — by identifying the structural skeleton common to every component and then locking that skeleton. This section records the dictionary, so the historical content remains attached to the abstract framework.

2.1 The BSDT pipeline in one paragraph

BSDT models N institutions with state $X \in \mathbb{R}^{N \times d}$, calibrates a normal-period Gaussian $\mathcal{N}(\mu_0, \Sigma_0)$, and at each step computes (i) a Mahalanobis camouflage energy E_{BSDT} , (ii) its Frobenius gradient norm (the *MFLS* signal), (iii) the alignment between this gradient and a physical restoring force $F = -\alpha \tilde{X}_\mu - LX$ (centred drift plus Laplacian coupling), and (iv) an adaptive damping γ^* trading detected blind-spot energy against an intervention cost θ .

2.2 Recognising the (S, G, F_{base}) triple

Reading off BSDT structurally:

- the centred state $\tilde{X} = X - \mathbf{1}\mu_0^\top$ plays the role of a feature map,
- the inverse calibration covariance Σ_0^{-1} plays the role of a positive-definite metric,
- the centred drift plus Laplacian force is a domain-specific nominal field.

With the identifications

$$S(X) := \text{vec}(\tilde{X}), \quad G := I_d \otimes \Sigma_0^{-1}, \quad F_{\text{base}}(X) := -\alpha \tilde{X}_\mu - LX,$$

the BSDT objects line up with the canonical objects exactly:

BSDT object	Canonical symbol	Identification
Calibration mean μ_0	target set $\{E = 0\}$	$S(X) = 0 \iff X = \mathbf{1}\mu_0^\top$
Centred state $\tilde{X}$	feature map $S(X)$	affine, so Jacobian $J \equiv I$
Inverse covariance Σ_0^{-1}	metric G	constant in Part I
Camouflage energy E_{BSDT}	$E = S^T G S$	$\ \tilde{X}\Sigma_0^{-1/2}\ _F^2$
Energy gradient $2\tilde{X}\Sigma_0^{-1}$	$g_X = 2J^T G S$	equal because $J = I$
MFLS signal $\ \nabla E\ _F$	$\ g_X\ $	locked Euclidean norm
Alignment $\cos \theta_t$	$\langle F, g_X \rangle / \ g_X\ ^2$	projection coefficient
Adaptive damping $\gamma^* = \frac{E_{\text{BSDT}}}{E_{\text{BSDT}} + \theta}$	$\gamma = \frac{E}{E + \theta}$	<i>identical formula</i>
Friction step on F	$\dot{X} = F - \gamma \text{proj}_{g_X}(F)$	locked canonical ODE

2.3 What the canonicalisation buys

Once the dictionary above is fixed, every theorem proved in Part I about the canonical ODE is automatically a theorem about the GravityEngine + BSDT loop:

- Lyapunov descent of E_{BSDT} (Theorem 9.1),
- asymptotic stability of the calibration centroid (Theorem 9.2),
- explicit exponential rate $\rho = 4\sigma^2\mu_G^2/M_G \cdot (1 - \gamma_{\max})$ (Theorem 9.3),
- local-in-energy ultimate boundedness with admissibility threshold (Theorem 9.4).

2.4 Where the v3 / v4 extensions enter the BSDT story

The three extensions are not departures from BSDT — they are the mathematical resolution of three concrete limitations of the original pipeline:

- **Multi-scale BSDT (Part II, fractal S).** BSDT’s four channels $(\delta_C, \delta_G, \delta_A, \delta_T)$ live at a single calibration scale; the fractal feature map S_{frac} stacks them across resolutions with Hausdorff-dimension weighting, recovering contagion modes between the single-window δ_C and the PCA-residual δ_G .
- **Conserved-direction BSDT (Part II, contact S).** When the system carries a conserved Reeb direction (calendar time, total assets), the contact feature map prevents γ^* from damping the conserved flow.
- **Time-varying calibration (Part II, Ricci-flow G).** Σ_0 in BSDT is fixed at calibration. Letting $G(t)$ evolve by a Ricci-type flow models a drifting normal-period geometry; Theorem F.2 states when re-calibration during stress is safe.
- **Fisher-optimal calibration (Part III).** Replacing Σ_0^{-1} by the feature-space Fisher information $\mathcal{I}_S$ yields the statistically optimal preconditioner; the Cramér–Rao remark makes precise the ceiling on any BSDT calibration’s descent rate.

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Historical anchor. Throughout the rest of this paper, every time a result is proved for the canonical kernel $\dot{X} = F - \gamma \text{proj}_{g_X}(F)$ with $\gamma = E/(E + \theta)$, that result specialises — under the dictionary of this section — to a statement about the GravityEngine + BSDT pipeline. The locked ODE is the abstract form of the BSDT damping law; nothing of the original is lost.

3 Foundations and Assumptions

This section fixes the regularity class, declares the standing assumptions, resolves the apparent singularity in the canonical ODE, and establishes well-posedness. Every later proof depends on the results stated here.

3.1 Standing assumptions

A1 – A4: Standing assumptions

(A1) Regularity of S :

$S \in C^2(\mathbb{R}^n, \mathbb{R}^k)$. The Jacobian $J(X) = \partial S / \partial X$ and the second derivative $\partial_X J$ are locally bounded.

(A2) Metric:

$G \in \mathbb{R}^{k \times k}$ is symmetric positive-definite and *constant* in X and t . We denote $0 < \mu_G \leq M_G$ for its smallest/largest eigenvalues. (*Relaxed in Parts II–III to (A2'):* $G \in C^1(\mathbb{R}^n, \text{Sym}_+^k)$ with uniform bounds $\mu_G I \preceq G(X) \preceq M_G I$. Every result of Part I is stated under (A2); Parts II–III state explicitly when (A2') replaces (A2).)

(A3) Nominal field:

$F_{\text{base}} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is locally Lipschitz.

(A4) Control parameter:

$\theta > 0$ is fixed.

3.2 The singular set and its resolution

The canonical ODE (Section 4) contains the factor $\|g_X\|^{-2}$ and is undefined on the *singular set* $\Sigma = \{X \in \mathbb{R}^n : g_X(X) = 0\}$.

Lemma 3.1 (Structure of Σ). *Under (A1)–(A2),*

$$g_X(X) = 0 \iff J(X)^T G S(X) = 0.$$

In particular:

- (i) If $\text{rank } J(X) = k$ (full row rank), then $g_X(X) = 0 \iff S(X) = 0 \iff E(X) = 0$.
- (ii) Outside the rank-deficient locus $\mathcal{R} = \{X : \text{rank } J(X) < k\}$, Σ coincides with the target set $\{E = 0\}$.

Proof. $g_X = 2J^T GS$, hence $g_X = 0 \iff J^T GS = 0$. If J has full row rank, J^T is injective on $\mathbb{R}^k$, so $GS = 0$, and since G is invertible, $S = 0$, giving $E = S^T GS = 0$. Outside $\mathcal{R}$ this holds at every point. $\square$

Regularised canonical ODE

$$\dot{X} = F_{\text{base}} - g_X - \gamma \frac{\langle F_{\text{base}} - g_X, g_X \rangle}{\|g_X\|^2 + \varepsilon} g_X, \quad \varepsilon \geq 0.$$

The *exact* canonical ODE corresponds to $\varepsilon = 0$. In implementation a small $\varepsilon > 0$ avoids division by zero; the limit $\varepsilon \rightarrow 0^+$ recovers the canonical equation away from Σ . By Lemma 3.1 the limit is benign whenever J has full row rank along trajectories.

3.3 Well-posedness

Theorem 3.2 (Local existence and uniqueness). *Assume (A1)–(A4) and let $X_0 \notin \Sigma$. There exists $T_* > 0$ and a unique C^1 solution $X : [0, T_*) \rightarrow \mathbb{R}^n$ of the canonical ODE with $X(0) = X_0$. If $\varepsilon > 0$, the same conclusion holds for every $X_0 \in \mathbb{R}^n$.*

Proof. Let $H(X) = F_{\text{base}}(X) - g_X(X) - \gamma(X)\alpha(X)g_X(X)$ with $\alpha(X) = \langle F_{\text{base}} - g_X, g_X \rangle / (\|g_X\|^2 + \varepsilon)$. Under (A1)–(A3) the maps $X \mapsto S, J, GS, g_X, E, \gamma$ are locally Lipschitz. On $\mathbb{R}^n \setminus \Sigma$ (or all of $\mathbb{R}^n$ if $\varepsilon > 0$), the denominator is bounded away from zero on a neighbourhood of X_0 , so α is locally Lipschitz, and so is H . Picard–Lindelöf yields the conclusion. $\square$

Corollary 3.3 (Global existence under energy bound). *If $\dot{E} \leq 0$ along the solution and the sublevel set $\{E \leq E(X_0)\}$ is compact, the solution is defined for all $t \geq 0$.*

4 Canonical Core Equation

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THE CANONICAL ODE (single, unique, not to be modified):

$$\dot{X} = F_{\text{base}}(X) - g_X(X) - \gamma(X) \frac{\langle F_{\text{base}}(X) - g_X(X), g_X(X) \rangle}{\|g_X(X)\|^2} g_X(X)$$

The three terms decompose cleanly:

- $F_{\text{base}}(X)$: domain-specific nominal vector field.
- $-g_X(X)$: geometric correction forcing E to decrease.
- $-\gamma(X)(\cdots)g_X(X)$: adaptive damping that removes the component of $F = F_{\text{base}} - g_X$ aligned with g_X , controlled by energy level.

5 Locked Definitions

All definitions below are *canonical and final*. No variants, no alternatives, no extensions are permitted inside the core system.

5.1 Representation map

A.1 — Representation

$$S : X \mapsto \mathbb{R}^k$$

$X \in \mathbb{R}^n$ is the state. $S(X) \in \mathbb{R}^k$ is the geometric feature vector. $k \geq 1$ is fixed for each domain.

The Jacobian of S at X is

$$J(X) = \frac{\partial S}{\partial X} \in \mathbb{R}^{k \times n}.$$

5.2 Energy functional

A.2 — Energy

$$E(X) = S(X)^T G S(X), \quad G \succ 0, \quad G \in \mathbb{R}^{k \times k}$$

G is symmetric positive-definite. $E(X) \geq 0$ with $E(X) = 0 \iff G^{1/2}S(X) = 0$.

5.3 State-space gradient

A.3 — Gradient (single, unique)

$$g_X(X) = J(X)^T \nabla_S E = 2 J(X)^T G S(X)$$

Derivation. By the chain rule, $\nabla_X E = J^T \nabla_S E$. Since $E = S^T G S$, one has $\nabla_S E = 2GS$. Therefore $g_X = J^T(2GS) = 2J^T GS$. $\square$

No other gradient definition exists in this system.

5.4 Modified dynamics

A.4 — Modified force

$$F(X) = F_{\text{base}}(X) - g_X(X)$$

F is the vector field *before* the adaptive projection term. The full dynamics (Section 4) add the projection of F onto $-g_X$.

5.5 Adaptive control gain

A.5 — Control gain

$$\gamma(X) = \frac{E(X)}{E(X) + \theta}, \quad \theta > 0$$

- $\gamma \in [0, 1)$ for all X .
- $\gamma \rightarrow 0$ as $E \rightarrow 0$: no damping near the target manifold.
- $\gamma \rightarrow 1$ as $E \rightarrow \infty$: full projection damping far away.
- θ is the only tunable scalar in the core system.

6 System Rules (Inviolable)

Rule 1 — Geometry location

Geometry enters *only* through g_X

The matrix G and the map S encode all geometric structure. Nothing else in the core ODE carries geometric information.

Rule 2 — Euclidean projection

All projections use the standard Euclidean inner product on $\mathbb{R}^n$

$$\langle u, v \rangle = u^T v, \quad \|u\| = \sqrt{u^T u}$$

There is no metric-weighted inner product in the dynamics layer.

Rule 3 — Pulled-back metric scope

$G_{\text{pull}} = J^T G J$ is used *only* for curvature (Hessian) analysis

G_{pull} does **not** appear in $\dot{X}$, in g_X , or in any projection.

Rule 4 – No duplication

Each object has exactly one definition; no parallel construction is permitted

In particular: there is no separate $\nabla_X \Phi$, no second force definition, and no alternative projection. The objects S , E , g_X , F , γ and $\dot{X}$ are defined exactly once, in Section 5, and used verbatim everywhere else.

Forbidden

The following are **permanently forbidden** in any version of this document:

- G_{pull}^{-1} appearing in the dynamics.
- Metric-weighted projections ($\langle u, v \rangle_G$ in the ODE).
- An independent $\nabla_X \Phi$ defined separately from g_X .
- A second or alternative definition of the force.
- Duplicate section headers or symbol re-definitions.

7 Forward Chain

The unique, linear data-flow through the system:

$$X \xrightarrow{S} S \xrightarrow{E=S^T G S} E \xrightarrow{2J^T G S} g_X \xrightarrow{\dot{X}=F_{\text{base}}-g_X-\gamma(\dots)g_X} \dot{X}$$

Each arrow is a unique map; no step branches or has an alternative.

§B.2 State-Space Dynamics

Setup. All dynamical quantities live in the state space $\mathbb{R}^n$. The representation map S and the energy E are scalar objects computed from X but the forces F_{base} , g_X , and F are all $\mathbb{R}^n$ -valued.

Projection. The projection of any vector $v \in \mathbb{R}^n$ onto g_X is

$$\text{proj}_{g_X}(v) = \frac{\langle v, g_X \rangle}{\|g_X\|^2} g_X,$$

computed with the *Euclidean* inner product. The complement is $v - \text{proj}_{g_X}(v)$.

Geometric content. The geometry of the target manifold/set is entirely captured by g_X . Equations of motion do not reference G , J , or S directly; they see only g_X computed from those quantities.

§B.2a Modified Force Definition

§B.2a — Modified Force (authoritative, single definition)

$$F(X) = F_{\text{base}}(X) - g_X(X)$$

The full canonical ODE (Section 4) is equivalently written as:

$$\dot{X} = F(X) - \gamma(X) \frac{\langle F(X), g_X(X) \rangle}{\|g_X(X)\|^2} g_X(X).$$

This removes from F its component along g_X , scaled by γ .

§C.5 Hessian and Curvature Tensor

This section derives the Hessian of E explicitly. The derivation introduces the second-derivative tensor of the representation map, which is the only tensorial object the system requires.

The representation curvature tensor K

C.5.0 — Representation curvature tensor

For $S \in C^2(\mathbb{R}^n, \mathbb{R}^k)$ define

$$K^i_{ab}(X) := \partial_a \partial_b S^i(X) = \partial_a J^i_b(X), \quad i \in \{1, \dots, k\}, \quad a, b \in \{1, \dots, n\}.$$

$K(X) \in \mathbb{R}^{k \times n \times n}$ is the third-order tensor of second partials of S . By Schwarz's theorem $K^i_{ab} = K^i_{ba}$; each *slice* $K^i \in \mathbb{R}^{n \times n}$ is symmetric.

Derivation of the Hessian of E

Proposition 7.1 (State-space Hessian, derived form). *Under (A1)–(A2), with index convention as above and Einstein summation,*

$$(\nabla_X^2 E)_{ab} = 2(J^T G J)_{ab} + \eta_i K^i_{ab}, \quad \eta := \nabla_S E = 2GS.$$

Equivalently, in coordinate-free form,

$$\nabla_X^2 E = 2G_{\text{pull}} + \langle \eta, K \rangle_1, \quad \langle \eta, K \rangle_1 := \sum_{i=1}^k \eta_i K^i.$$

Proof. Write $E = G_{ij} S^i S^j$ with $G^T = G$. Then

$$\begin{aligned} \partial_a E &= G_{ij} (\partial_a S^i) S^j + G_{ij} S^i (\partial_a S^j) \\ &= 2G_{ij} J^i_a S^j = 2(J^T G S)_a = (g_X)_a, \end{aligned}$$

recovering $\nabla_X E = g_X$. Differentiating once more,

$$\begin{aligned} \partial_b \partial_a E &= 2G_{ij} ((\partial_b J^i_a) S^j + J^i_a (\partial_b S^j)) \\ &= 2G_{ij} K^i_{ab} S^j + 2G_{ij} J^i_a J^j_b \\ &= \underbrace{(2GS)_i K^i_{ab}}_{=\eta_i K^i_{ab}} + 2(J^T G J)_{ab}. \end{aligned}$$

Symmetry in (a, b) holds because K^i and $J^T G J$ are symmetric. $\square$

Remark 7.2 (Two ingredients). The state-space Hessian decomposes uniquely into:

- the **pulled-back metric** $2G_{\text{pull}} = 2J^T G J$, capturing the linear part of the geometry;
- the **curvature contraction** $\langle \eta, K \rangle_1$, capturing how the representation map S itself bends.

The curvature contraction vanishes whenever S is affine ($K \equiv 0$) or when $\eta = 2GS = 0$ (i.e. on the target set $\{E = 0\}$). Hence on the target set, $\nabla_X^2 E|_{\{E=0\}} = 2G_{\text{pull}}$ exactly.

Remark 7.3 (Operator norm). Define the tensor norm

$$\|K\|_{\text{op}} := \sup_{\|u\|=\|v\|=1, \|w\|=1} \sum_{i,a,b} K^i_{ab} u_i v_a w_b.$$

Then the curvature contraction obeys $\|\langle \eta, K \rangle_1\|_2 \leq \|K\|_{\text{op}} \|\eta\|_2$, which is the bound used in Proposition 7.6.

Remark 7.4 (Role in the dynamics). $G_{\text{pull}} = J^T G J$ and K both appear here as part of the Hessian. By Rule 3, neither appears in $\dot{X}$ or g_X .

§D.3 Curvature Manifold

§D.3 — Curvature manifold

$$\mathcal{C}_{\text{man}} = \{X \in \mathbb{R}^n : \lambda_{\max}(\nabla_X^2 E(X)) = 1\}$$

$\mathcal{C}_{\text{man}}$ is the level set of maximal principal curvature equal to 1. It separates regions where the energy landscape is convex-dominated ($\lambda_{\max} < 1$) from regions where nonlinear curvature terms dominate ($\lambda_{\max} > 1$).

Proposition 7.5. $\mathcal{C}_{\text{man}}$ is generically a smooth $(n-1)$ -dimensional submanifold of $\mathbb{R}^n$ wherever $\lambda_{\max}$ is a simple eigenvalue and its derivative is non-zero.

Proof. Let $h(X) = \lambda_{\max}(\nabla_X^2 E(X)) - 1$ so that $\mathcal{C}_{\text{man}} = h^{-1}(0)$. By (A1) the entries of $\nabla_X^2 E$ are continuous; on the open set where $\lambda_{\max}$ is simple, classical perturbation theory (Kato) gives $\lambda_{\max} \in C^1$ with derivative $\nabla h = v^T (\partial_X \nabla_X^2 E) v$, where v is the unit eigenvector. Wherever $\nabla h \neq 0$ the implicit function theorem yields the smooth codimension-1 structure. $\square$

Proposition 7.6 (Curvature bound). Under (A1)–(A2), with $\|K\|_{\text{op}}$ as defined in Section 7,

$$\lambda_{\max}(\nabla_X^2 E) \leq 2M_G \|J\|_2^2 + \|K\|_{\text{op}} \|\eta\|_2, \quad \eta = 2GS.$$

In particular, on $\mathcal{C}_{\text{man}}$, $2M_G \|J\|_2^2 + \|K\|_{\text{op}} \|2GS\|_2 \geq 1$.

Proof. By Proposition 7.1, $\nabla_X^2 E = 2J^T G J + \langle \eta, K \rangle_1$. The spectral norm is sub-additive; bound the first term using $\|J^T G J\|_2 \leq \|G\|_2 \|J\|_2^2 \leq M_G \|J\|_2^2$, and the second using $\|\langle \eta, K \rangle_1\|_2 \leq \|K\|_{\text{op}} \|\eta\|_2$ from Section 7. $\square$

The practical role of $\mathcal{C}_{\text{man}}$:

- Provides a natural domain boundary for local linearisation arguments.
- Guides the choice of θ in γ : setting θ so that $\gamma \approx \frac{1}{2}$ on $\mathcal{C}_{\text{man}}$ balances correction and tracking.

§H.1 Numerical Implementation

H.1.1 Per-step algorithm

Given state X_t , advance one step:

1. Compute $S \leftarrow S(X_t)$.
2. Compute $J \leftarrow \partial S / \partial X|_{X_t}$.

3. Compute $g_X \leftarrow 2J^T GS$. (gradient — **single computation**)
4. Compute $E \leftarrow S^T GS$. (energy)
5. Compute $F_{\text{base}} \leftarrow F_{\text{base}}(X_t)$.
6. Compute $F \leftarrow F_{\text{base}} - g_X$. (modified force — **not** F_{base} alone)
7. Compute $\gamma \leftarrow E/(E + \theta)$.
8. Compute projection coefficient $\alpha \leftarrow \langle F, g_X \rangle / \|g_X\|^2$.
9. Set $\dot{X} \leftarrow F - \gamma \alpha g_X$. (canonical ODE)
10. Integrate: $X_{t+h} \leftarrow X_t + h \dot{X}$ (or higher-order scheme).
11. **Termination:** stop when $E < \varepsilon_E$ and $\|g_X\| < \varepsilon_g$ for τ consecutive steps (cf. §H.1.4).

Forbidden

Step 6 uses $F = F_{\text{base}} - g_X$, not F_{base} alone. Step 8 uses this F in the projection, not F_{base} .

H.1.2 Gradient correctness check

Before running any simulation, verify:

$$g_X \stackrel{?}{=} 2J^T GS,$$

by finite-difference: $\Delta E / \Delta X_i$ should match $(g_X)_i$ to machine precision for small perturbations.

H.1.3 Energy monotonicity (derivation)

Lemma 7.7 (Energy derivative). *Along any solution of the canonical ODE,*

$$\dot{E} = \langle g_X, \dot{X} \rangle = (1 - \gamma) (\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2).$$

Equivalently, with $F = F_{\text{base}} - g_X$ and $\alpha = \langle F, g_X \rangle / \|g_X\|^2$,

$$\dot{E} = \langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 - \gamma \alpha \|g_X\|^2 = \langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 - \gamma \langle F, g_X \rangle.$$

Proof. Differentiate $E = S^T GS$: $\dot{E} = 2S^T G \dot{S} = 2S^T G J \dot{X} = (2J^T GS)^T \dot{X} = \langle g_X, \dot{X} \rangle$. Substitute the canonical ODE $\dot{X} = F - \gamma \alpha g_X$ to get $\dot{E} = \langle g_X, F \rangle - \gamma \alpha \|g_X\|^2 = \langle g_X, F \rangle - \gamma \langle F, g_X \rangle = (1 - \gamma) \langle F, g_X \rangle$. Since $\langle F, g_X \rangle = \langle F_{\text{base}}, g_X \rangle - \|g_X\|^2$, the two displayed forms follow. $\square$

Pure geometric flow ($F_{\text{base}} = 0$). Then $F = -g_X$, so

$$\dot{E} = -(1 - \gamma) \|g_X\|^2 \leq 0,$$

strictly negative whenever $g_X \neq 0$ (and $\gamma < 1$, always).

Energy descent condition. Since $1 - \gamma > 0$, $\dot{E} \leq 0$ holds if and only if

$$\langle g_X, F_{\text{base}} \rangle \leq \|g_X\|^2.$$

This is the explicit angle condition between F_{base} and g_X used in the stability theorems of Section 9.

H.1.4 Step-size, stopping, and Jacobian computation

Stopping criterion. Halt when both $E(X_t) < \varepsilon_E$ and $\|g_X(X_t)\| < \varepsilon_g$ hold for τ consecutive steps (dwell time), with default tolerances $\varepsilon_E = 10^{-8}$, $\varepsilon_g = 10^{-6}$, $\tau = 5$.

Adaptive step-size. Use $h_t = h_0 / (1 + \lambda_{\max}(\nabla_X^2 E(X_t)))$, so $h_t \leq h_0$ everywhere and $h_t \rightarrow h_0/2$ on $\mathcal{C}_{\text{man}}$. By Proposition 7.6, $\lambda_{\max}$ is bounded in terms of $\|J\|$, $\|G\|$, and the curvature of S , so h_t is uniformly positive on bounded sets.

Choice of G . The conditioning of $g_X = 2J^T GS$ scales with $\kappa(G)$. Default rule: choose G diagonal with $G_{ii} = 1/\widehat{\text{Var}}(S_i)$ from data (or design), so E becomes the squared Mahalanobis norm of S and the energy landscape is approximately isotropic in the operating region.

Jacobian computation. In order of preference: (i) closed-form analytic J ; (ii) automatic differentiation when S is given as code; (iii) finite differences with step $\delta = \sqrt{\epsilon_{\text{mach}}}$ at cost $O(nk)$ evaluations per step.

8 Predictive scalars: direction and momentum

The locked ODE is first-order, so the system has no inertial state $\dot{X}$ that could be tracked. *Direction* and *momentum* must therefore be read off scalar observables built from the same quantities $(S, g_X, F_{\text{base}}, E, \gamma)$ that already drive the dynamics. This section collects the canonical predictors. They are all chart-dependent in Part I; their Fisher (chart-covariant) version appears in Part III.

8.1 Direction predictor: alignment $\cos \theta_t$

Alignment scalar

$$\cos \theta_t := \frac{\langle F_{\text{base}}(X_t), g_X(X_t) \rangle}{\|F_{\text{base}}(X_t)\| \|g_X(X_t)\|} \in [-1, +1].$$

The *sign* of $\cos \theta_t$ is the binary direction indicator (toward target / away); the *magnitude* measures how much of F_{base} acts radially on the energy.

- $\cos \theta_t \rightarrow +1$: nominal field and energy gradient are co-aligned — coherent collapse pressure.
- $\cos \theta_t \rightarrow 0$: F_{base} is tangent to the level set $\{E = \text{const}\}$ — circulation, no radial progress; direction undetermined by energy alone.
- $\cos \theta_t \rightarrow -1$: anti-aligned — restoring competition, E falls without help from F_{base} .

8.2 Momentum predictors: $\dot{E}$, ρ_{eff} , MFLS, γ

Differentiating $E = S^T G S$ along the canonical ODE and using $g_X = 2J^T G S$ gives the closed expression

$$(1) \quad \dot{E}(t) = \langle g_X, F_{\text{base}} \rangle - \gamma(X_t) \frac{\langle F_{\text{base}}, g_X \rangle^2}{\|g_X\|^2}.$$

Equation (1) is the master momentum identity: it is the quantity Theorems 9.1–9.3 bound from above. Four scalars are read off it.

Momentum scalars

- Energy velocity** $v_E(t) := -\dot{E}(t)$. Positive v_E means the system is contracting toward the target; magnitude is the instantaneous descent rate.
- Effective decay rate** $\rho_{\text{eff}}(t) := \frac{-\dot{E}(t)}{E(t)}$. Predicts the local exponential half-life $\tau_{1/2}(t) = \ln 2 / \rho_{\text{eff}}(t)$.
- Gradient capacity (MFLS)** $\|g_X\| = 2\|J^T G S\|$. Upper-bounds $|\dot{E}|$ via Cauchy–Schwarz and equals it when $|\cos \theta_t| = 1$ and $\gamma = 0$.
- Damping commitment** $\gamma = E/(E + \theta) \in [0, 1)$. $\gamma \rightarrow 1$ flags maximum projection load.

Leading vs. lagging. $\cos \theta_t$ and $\rho_{\text{eff}}(t)$ are *leading* — they move before the state visibly does, because they depend on g_X and F_{base} , not just X . $E(t)$ and $\gamma(X_t)$ are *lagging* — they integrate the leading signals.

8.3 Compact dashboard

The minimal predictive log per step is the quintuple

$$\left(\cos \theta_t, \|g_X(X_t)\|, E(X_t), \gamma(X_t), \rho_{\text{eff}}(t) \right).$$

Their roles:

Symbol	Reads as	Dir.	Mom.	Lead / lag
$\cos \theta_t$	alignment	✓ sign+tilt	—	leading
$\ g_X\ $ (MFLS)	gradient capacity	—	✓ cap.	leading
E	Lyapunov energy	✓ closeness	via $\dot{E}$	lagging
γ	damping commitment	—	✓	lagging
ρ_{eff}	local decay rate	—	✓ predicted $\tau_{1/2}$	leading

8.4 Predictive use of the main theorems

The stability theorems of Section 9 convert the above scalars into *guaranteed* bounds:

- Theorem 9.1 guarantees $\dot{E} \leq 0$ along the canonical ODE, so $\rho_{\text{eff}}(t) \geq 0$ whenever $F_{\text{base}} \equiv 0$.
- Theorem 9.3 furnishes a lower bound $\rho_{\text{eff}}(t) \geq \rho = 4\sigma^2 \mu_G^2 / M_G (1 - \gamma_{\text{max}})$ under the uniform Jacobian rank condition. The ratio $\rho_{\text{eff}}(t)/\rho$ is therefore a running *efficiency score*: values $\gg 1$ indicate the instantaneous regime is far better than the worst-case guarantee; values $\rightarrow 1$ signal saturation of the bound.
- Theorem 9.4 caps how large E may settle when $F_{\text{base}} \neq 0$: a sustained drift of E above the admissibility threshold Ψ^* is the early-warning signal that the local-in-energy capture window has been exited.

8.5 Information-geometric upgrade (preview of Part III)

Under the Fisher choice $G(X) = \mathcal{I}_S(S(X))$ (Section H) the predictors $\cos \theta_t$ and ρ_{eff} become chart-covariant in the auxiliary natural-gradient flow (Theorem I.1), and the Cramér–Rao remark places a statistical ceiling on ρ_{eff} . The ratio

$$\eta_{\text{cal}}(t) := \frac{\rho_{\text{eff}}(t)}{\rho_{\text{CR}}}$$

is then a *calibration-efficiency* signal: values close to 1 mean the current G is statistically optimal; values $\ll 1$ flag that the metric should be re-estimated. This is the v4 meta-momentum indicator.

Forbidden

None of the scalars in this section may be substituted into the locked ODE: the dynamics use g_X , γ , F_{base} exactly as defined in Sections 4–6. The predictive scalars are *observables*, not control inputs.

9 Main Stability Theorems

We now state the principal results of the paper. Throughout this section (A1)–(A4) hold and the trajectory $X(\cdot)$ is a solution of the canonical ODE on a maximal interval of existence.

9.1 Lyapunov descent

Theorem 9.1 (Lyapunov descent). *For any solution $X(\cdot)$ on $\mathbb{R}^n \setminus \Sigma$,*

$$\dot{E}(t) = (1 - \gamma) (\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2).$$

In particular:

- (i) If $F_{\text{base}} \equiv 0$, then $\dot{E} = -(1 - \gamma)\|g_X\|^2 < 0$ whenever $g_X \neq 0$.
- (ii) In general, $\dot{E} \leq 0$ iff $\langle g_X, F_{\text{base}} \rangle \leq \|g_X\|^2$.

Proof. Direct from Lemma 7.7; (i) is the special case $F_{\text{base}} = 0$, and (ii) follows since $1 - \gamma > 0$. $\square$

9.2 Asymptotic stability

Theorem 9.2 (Asymptotic stability of the target set). *Assume (A1)–(A4), $F_{\text{base}} \equiv 0$, and that $J(X)$ has full row rank on every sublevel set $\{E \leq c\}$ that is compact. Then for every initial condition X_0 with $\{E \leq E(X_0)\}$ compact, the solution exists for all $t \geq 0$ and*

$$E(X(t)) \xrightarrow[t \rightarrow \infty]{} 0, \quad \text{dist}(X(t), \{E = 0\}) \rightarrow 0.$$

Proof. By Theorem 9.1(i) $\dot{E} \leq 0$, so $X(t)$ stays in the compact sublevel set; Corollary 3.3 gives global existence. LaSalle's invariance principle implies $X(t)$ converges to the largest invariant set inside $\{\dot{E} = 0\} = \{g_X = 0\}$. By Lemma 3.1(i) the rank assumption forces $\{g_X = 0\} = \{E = 0\}$, which is itself invariant (since the right-hand side vanishes there). Hence $E(X(t)) \rightarrow 0$. $\square$

9.3 Exponential rate

Theorem 9.3 (Exponential convergence). *Assume the hypotheses of Theorem 9.2 and that on the invariant compact set there exists $\sigma > 0$ with $\sigma_{\min}(J(X)) \geq \sigma$. Then*

$$E(X(t)) \leq E(X_0) e^{-\rho t}, \quad \rho = 4\sigma^2 \frac{\mu_G^2}{M_G} (1 - \gamma_{\max}),$$

where $\gamma_{\max} = E(X_0)/(E(X_0) + \theta) < 1$.

Proof. With $F_{\text{base}} = 0$, $\dot{E} = -(1 - \gamma)\|g_X\|^2$. We bound $\|g_X\|^2 = 4\|J^T GS\|^2 \geq 4\sigma^2\|GS\|^2 \geq 4\sigma^2\mu_G^2\|S\|^2$, while $E = S^T GS \leq M_G\|S\|^2$, giving $\|S\|^2 \geq E/M_G$. Hence

$$\|g_X\|^2 \geq \frac{4\sigma^2\mu_G^2}{M_G} E.$$

With $\gamma \leq \gamma_{\max} < 1$ monotone in E , $\dot{E} \leq -\rho E$ and Grönwall's inequality concludes. $\square$

9.4 Ultimate boundedness with non-zero F_{base}

Theorem 9.4 (Ultimate boundedness). *Assume (A1)–(A4), the rank condition of Theorem 9.3, and that $\|F_{\text{base}}\| \leq M$ on the operating region. Define*

$$\Psi(R) := \frac{\theta}{R + \theta} \cdot 2\sigma \sqrt{\frac{\mu_G^2}{M_G}} R = (1 - \gamma(R)) \cdot 2\sigma \frac{\mu_G}{\sqrt{M_G}} \sqrt{R}.$$

Then Ψ is unimodal on $(0, \infty)$ with maximum $\Psi^* = \sigma\theta\mu_G/\sqrt{M_G}$ attained at $R^* = \theta$. Assuming the admissibility condition $M < \Psi^*$, the equation $\Psi(R) = M$ has two roots $0 < R_- \leq R_+$. If the trajectory enters the admissible energy window $E(X(t_0)) \leq R_+$, then for any $R \in [R_-, R_+]$, $E(X(t)) \leq R$ for all sufficiently large $t \geq T(R, X(t_0))$. In particular, the theorem is a local-in-energy ultimate-boundedness result on the operating window $\{E \leq R_+\}$; it does not claim global capture from arbitrarily large energies.

Proof. From Lemma 7.7, $\dot{E} = (1 - \gamma)(\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2) \leq -(1 - \gamma)\|g_X\|(\|g_X\| - M)$. On $\{E \geq R\}$ the rank condition and the bound from Theorem 9.3 give $\|g_X\|^2 \geq 4\sigma^2(\mu_G^2/M_G)R$, so $(1 - \gamma(R))\|g_X\| \geq \Psi(R)$. For $R \in [R_-, R_+]$, $\Psi(r) \geq M$ for every $r \in [R, R_+]$. Therefore $\dot{E} \leq 0$

on the annulus $\{R \leq E \leq R_+\}$, so once a trajectory is in the operating window $\{E \leq R_+\}$ it cannot cross upward through R_+ and is pulled into $\{E \leq R\}$ in finite time. $\square$

Remark 9.5 (Critical disturbance). The admissibility threshold $\Psi^* = \sigma\theta\mu_G/\sqrt{M_G}$ is the largest disturbance amplitude M for which any sublevel set is forward-invariant. For $M \geq \Psi^*$, no finite ultimate bound is provided by the canonical descent alone; an outer feedback shaping F_{base} is required.

9.5 Long–short symmetry of the locked ODE

The Lyapunov function $E = S^T G S$ is even in the feature S . The canonical ODE inherits an exact $\mathbb{Z}_2$ symmetry that formalises the long/short equivalence used in futures trading (Section 10).

Theorem 9.6 ($\mathbb{Z}_2$ symmetry). *Suppose S admits an involutive lift $X \mapsto X^\sharp$ with $S(X^\sharp) = -S(X)$, $J(X^\sharp) = -J(X) dX^\sharp/dX$, and $F_{\text{base}}(X^\sharp) = -F_{\text{base}}(X)$. Then under $X(t) \mapsto X^\sharp(t)$:*

$$E \rightarrow E, \quad g_X \rightarrow -g_X, \quad \gamma \rightarrow \gamma, \quad \dot{X} \rightarrow -\dot{X}.$$

In particular, every trajectory of the canonical ODE has a mirror trajectory under sign reversal, and the two converge to $\{E = 0\}$ at the same rate $\rho_{\text{eff}}(t)$.

Proof. Direct: E is quadratic in S , so $E(X^\sharp) = S^T G S = E(X)$. For the gradient, $g_X(X^\sharp) = 2J(X^\sharp)^T G S(X^\sharp) = 2(-J)^T G (-S) = 2J^T G S = g_X(X)$ when the lift acts trivially on the state, and $-g_X(X)$ when it inverts the state coordinate; the latter is the operative case for futures (see §10). The damping $\gamma = E/(E + \theta)$ depends on E only, hence invariant. Substituting into the canonical ODE gives $\dot{X}^\sharp = -\dot{X}$. $\square$

Remark 9.7 (Operational reading for futures). For signed futures positions $X \in \mathbb{R}^n$ with F_{base} a forecast field, the canonical sizing $\Delta X = F_{\text{base}} - \gamma \langle F_{\text{base}}, g_X \rangle / \|g_X\|^2 g_X$ is automatically long/short symmetric: the sign of the trade is $\text{sgn} \langle F_{\text{base}}, -g_X \rangle$ and reverses with S . A single calibration of (G, θ) serves both directions. The two-sided admissibility band $-\sqrt{\Psi^*/\lambda_{\min}(G)} \leq S \leq \sqrt{\Psi^*/\lambda_{\min}(G)}$ is the canonical stop-out, symmetric in long and short.

10 Application Domains

Each subsection instantiates (S, G, F_{base}) for a specific domain, verifies (A1)–(A4), and cites the appropriate stability theorem from Section 9. *No structural change to the canonical ODE occurs.*

10.1 Trading / statistical-arbitrage spread control

Let $X = w \in \mathbb{R}^n$ be portfolio weights, $A \in \mathbb{R}^{k \times n}$ a factor-loading matrix, $b \in \mathbb{R}^k$ a target spread vector, and $\Sigma \in \mathbb{R}^{k \times k}$ the (PD) factor covariance.

$$S(w) = Aw - b, \quad G = \Sigma^{-1}, \quad F_{\text{base}}(w) = -\kappa(w - w^*),$$

with $\kappa > 0$ and w^* the alpha-target weight. Then $J = A$ (constant), $E(w) = (Aw - b)^T \Sigma^{-1} (Aw - b)$ is the squared Mahalanobis spread, and $g_X = 2A^T \Sigma^{-1} (Aw - b)$.

Corollary 10.1. *If A has full row rank, Theorem 9.3 applies with $\sigma = \sigma_{\min}(A)$ and gives exponential decay of the Mahalanobis spread; Theorem 9.4 bounds the long-run spread in terms of the alpha-tracking torque $\kappa \|w_0 - w^*\|$.*

Transaction costs enter via a Lipschitz extension of F_{base} (smooth proximal regularisation), preserving (A3).

10.2 Spectral / Galerkin PDE control

Let $X = (a_1, \dots, a_N) \in \mathbb{R}^N$ be the Galerkin coefficients of a target field, and a^* a desired spectrum.

$$S(X) = X - a^*, \quad G = \text{diag}(g_1, \dots, g_N), \quad F_{\text{base}}(X) = \mathcal{L}_N X,$$

where $g_k > 0$ are mode weights (e.g. $g_k = k^2$ for an H^1 -type energy) and $\mathcal{L}_N$ is the discretised PDE operator. Then $J = I_N$, $E = \sum_k g_k (a_k - a_k^*)^2$, and $g_X = 2G(X - a^*)$.

Corollary 10.2. *The rank condition is automatic ($J = I_N$). Theorem 9.3 gives convergence rate $\rho = 4\mu_G(1 - \gamma_{\max})$ in the weighted ℓ^2 norm. Stability of the underlying PDE in the full space requires $\sup_N \|\mathcal{L}_N\| < \infty$ in the weighted norm; under that uniform bound, the discrete results lift to the continuum limit.*

10.3 First-order optimisation

Minimise $f \in C^2(\mathbb{R}^n)$ with locally bounded Hessian. Set

$$S(X) = \nabla f(X), \quad G = I_n, \quad F_{\text{base}}(X) = -\nabla f(X).$$

Then $J(X) = \nabla^2 f(X)$, $E(X) = \|\nabla f(X)\|^2$, and $g_X = 2(\nabla^2 f) \nabla f$. The canonical ODE becomes a damped, projection-corrected gradient flow that vanishes at stationary points.

Corollary 10.3. *If $\nabla^2 f$ is uniformly positive-definite on a sublevel set of f , then $\sigma_{\min}(J) \geq \sigma > 0$ on that set, Theorem 9.3 applies, and $\|\nabla f(X(t))\|^2$ decays exponentially.*

Comparison with classical preconditioned descent: the canonical ODE adds the γ -modulated projection term, which bounds the transient overshoot of E at the cost of one extra inner product per step.

10.4 Task-space robotics control

Let $X = q \in \mathbb{R}^n$ be joint angles, $\text{FK} : \mathbb{R}^n \rightarrow \mathbb{R}^k$ the forward kinematics map, $p^* \in \mathbb{R}^k$ the desired end-effector pose, and $W \succ 0$ a task-space weight.

$$S(q) = \text{FK}(q) - p^*, \quad G = W, \quad F_{\text{base}}(q) = -K_p J(q)^T W (\text{FK}(q) - p^*),$$

with $K_p > 0$. Note $F_{\text{base}} = -\frac{K_p}{2} g_X$, so $F = F_{\text{base}} - g_X = -(K_p/2 + 1)g_X$ is fully aligned with $-g_X$, the projection coefficient becomes identically 1, and the canonical ODE collapses to scalar-damped descent $\dot{X} = -(1 + K_p/2)(1 - \gamma)g_X$. This domain therefore illustrates a degenerate alignment regime; richer behaviour arises when F_{base} is replaced by a state-feedback controller not collinear with g_X .

Corollary 10.4. *Away from kinematic singularities ($\text{rank } J = k$) the rank condition holds and Theorem 9.3 gives exponential pose convergence. Near singularities use the regularised ODE ($\varepsilon > 0$); under the regularisation the conclusion of Theorem 9.2 weakens to convergence to the closure of the invariant set $\{S \in \ker J^T\}$.*

11 Domain Parameterisation

LOCKED

Domain independence of the core system:

$$\boxed{\text{Different domains change only } S(X), \quad E(X), \quad F_{\text{base}}(X).}$$

The ODE structure, the gradient formula, the projection, and the control gain are unchanged across all domains.

Domain	$S(X)$	$E(X)$	$F_{\text{base}}(X)$
Trading / Signal	Spread / factor vector	$S^T G S$ with $G = \text{diag}(\sigma^{-2})$	Alpha signal + mean reversion
PDE control	Modal coefficients a_k	$\ a\ _G^2$ (weighted L^2)	PDE right-hand side
Optimisation	Gradient / constraint residual	$\ \nabla f\ ^2$ or barrier term	Gradient descent step
Robotics	Task-space error	$e^T W e$	Joint velocity command

Remark 11.1. The table entries are *examples*, not definitions. Any choice of S , G , and F_{base} satisfying the structural requirements of Sections 5.1–5.5 produces a valid instance of the canonical system.

12 Verification Checklist

All items below must evaluate to **TRUE** before any application or proof proceeds. If any item is **FALSE**, return to the indicated section and fix it.

Verification checklist		
✓ 1. Forward chain intact	$X \rightarrow S \rightarrow E \rightarrow g_X \rightarrow \dot{X}$ is the unique data path.	→ §7
✓ 2. Single gradient	$g_X = 2J^T G S$ is the only gradient appearing anywhere.	→ §5.3
✓ 3. Single force definition	$F = F_{\text{base}} - g_X$; no other force is named or used.	→ §7
✓ 4. Euclidean projection only	$\langle \cdot, \cdot \rangle = u^T v$; no $\langle \cdot, \cdot \rangle_G$ in dynamics.	→ Rule 2
✓ 5. No metric mixing	G_{pull} absent from $\dot{X}$ and g_X .	→ Rule 3
✓ 6. Control uses F, not F_{base}	Projection term $\langle F, g_X \rangle$; not $\langle F_{\text{base}}, g_X \rangle$.	→ §4
✓ 7. Hessian formula correct	$\nabla_X^2 E = 2J^T G J + \langle \eta, K \rangle_1$ with $\eta = 2GS$ and $K^i_{ab} = \partial_a \partial_b S^i$.	→ §C.5
✓ 8. Curvature manifold defined	$\mathcal{C}_{\text{man}} = \{X : \lambda_{\max}(\nabla_X^2 E) = 1\}$.	→ §D.3
✓ 9. Domain variation confined	Only S , E , F_{base} differ across domains.	→ §11
✓ 10. No structural changes after this version	Applications, proofs, and implementations only.	→ §13

13 Structure Freeze

LOCKED
NO further structural changes to this system. The core ODE (§4), the five definitions (§5), and the four rules (§6) are frozen. Permitted after this point:

- **Applications** — instantiate S, G, F_{base} for a new domain.
- **Proofs** — stability, convergence, curvature bounds.
- **Implementations** — numerical integrators, code, experiments.

Not permitted:

- Redefining g_X, F, γ , or the ODE.
- Adding metric-weighted projections.
- Introducing alternative gradient or force terms.

14 Summary — One Line

Lock system $\longrightarrow$ Build applications $\longrightarrow$ Prove results

A Symbol Table

Symbol	Meaning	Dimensions
X	State vector	$\mathbb{R}^n$
$S(X)$	Representation / feature map	$\mathbb{R}^k$
$J(X)$	Jacobian $\partial S / \partial X$	$\mathbb{R}^{k \times n}$
G	Feature-space metric (SPD, fixed)	$\mathbb{R}^{k \times k}$
$E(X)$	Energy $= S^T G S$	$\mathbb{R}_{\geq 0}$
$g_X(X)$	State-space gradient $= 2J^T G S$	$\mathbb{R}^n$
$F_{\text{base}}(X)$	Domain-specific nominal field	$\mathbb{R}^n$
$F(X)$	Modified force $= F_{\text{base}} - g_X$	$\mathbb{R}^n$
$\gamma(X)$	Adaptive gain $= E / (E + \theta)$	$[0, 1)$
θ	Gain regularisation constant	$\mathbb{R}_{> 0}$
G_{pull}	Pulled-back metric $= J^T G J$	$\mathbb{R}^{n \times n}$
$K(X)$	Curvature tensor of S , $K^i_{ab} = \partial_a \partial_b S^i$	$\mathbb{R}^{k \times n \times n}$
$\eta(X)$	Feature-gradient $= \nabla_S E = 2GS$	$\mathbb{R}^k$
$\mathcal{C}_{\text{man}}$	Curvature manifold $\lambda_{\text{max}} = 1$	$\subset \mathbb{R}^n$

B Quick Reference Card

Quantity	Formula
Representation	$S : X \rightarrow \mathbb{R}^k$
Energy	$E = S^T G S, \quad G \succ 0$
Gradient	$g_X = 2J^T G S$
Modified force	$F = F_{\text{base}} - g_X$
Gain	$\gamma = E / (E + \theta)$
Canonical ODE	$\dot{X} = F_{\text{base}} - g_X - \gamma \frac{\langle F, g_X \rangle}{\ g_X\ ^2} g_X$
Hessian	$\nabla_X^2 E = 2J^T G J + \langle \eta, K \rangle_1, \quad \eta = 2GS$
Curvature manifold	$\mathcal{C}_{\text{man}} = \{X : \lambda_{\max}(\nabla_X^2 E) = 1\}$

C Proof Roadmap (Next Steps)

The following results remain to be established (proofs are outside the locked core but follow directly from it):

- P1. Lyapunov descent.** *Done* – Theorem 9.1.
- P2. Invariant set / asymptotic stability.** *Done* – Theorem 9.2.
- P3. Curvature bound.** *Done* – Proposition 7.6.
- P4. Domain convergence.** *Done* – corollaries in Section 10.
- P5. Discretisation error.** *Open.* Bound integration error growth for Euler and Runge–Kutta schemes as a function of h and θ ; quantify the bias introduced by $\varepsilon > 0$ regularisation.
- P6. Stochastic extension.** *Open.* Itô analysis of $dX = (\cdots) dt + \sigma dW$, including the Itô correction to $\dot{E}$.
- P7. Time-varying $G(t)$.** *Open.* Effect on $\dot{E}$ and on the rate ρ of Theorem 9.3.

Part 1. Geometric Extensions

LOCKED

Extension principle (v3). Each extension below adds new geometric structure *strictly upstream* of g_X , by changing either the representation map S , the metric G , or the energy template. **The canonical ODE, the gradient formula $g_X = 2J^T G S$, and the Euclidean projection are unchanged.** The frozen v2 system from Part I is the *kernel* of every extension.

D Fractal-Geometric Extension

D.1 Motivation and setup

Many target sets are not smooth manifolds but *self-similar* attractors with non-integer Hausdorff dimension $d_H \in (0, n)$: financial support/resistance, turbulent attractors, multi-scale boundary

layers. The frozen core requires only that the set be expressed as the zero level set of a smooth energy. We therefore route all fractal information through S .

Let $\mathcal{A} \subset \mathbb{R}^n$ be a (possibly fractal) compact target and fix $\delta_0 > 0$. Define the δ -mollified distance

$$d_\delta(X) := \inf_{Y \in \mathcal{A}_\delta} \|X - Y\|, \quad \mathcal{A}_\delta := \{Y : \text{dist}(Y, \mathcal{A}) \leq \delta\}.$$

For $\delta > 0$, $\mathcal{A}_\delta$ is a smooth tubular thickening, so $d_\delta \in C^2(\mathbb{R}^n \setminus \mathcal{A}_\delta)$.

D.2 Multi-scale representation

F.1 — Fractal feature map

Fix scales $\delta_1 > \delta_2 > \dots > \delta_L > 0$ and define

$$S_{\text{frac}}(X) := (d_{\delta_1}(X), \dots, d_{\delta_L}(X))^T \in \mathbb{R}^L.$$

The fractal metric is $G_{\text{frac}} = \text{diag}(\delta_1^{-2d_H}, \dots, \delta_L^{-2d_H})$.

Why this metric. For a d_H -dimensional self-similar set, $\text{Vol}(\mathcal{A}_\delta) \sim \delta^{n-d_H}$ and the typical squared distance at scale δ scales as δ^2 . The weight $\delta_\ell^{-2d_H}$ rescales the contribution of each shell so that mass is balanced across scales according to the Hausdorff dimension. Strict scale-invariance under simultaneous dilation $\mathcal{A} \mapsto \lambda\mathcal{A}$, $\delta \mapsto \lambda\delta$ (where each summand transforms as λ^{2-2d_H}) holds only at $d_H = 1$; for general d_H the weight produces *dimension-aware* but not invariant weighting, which is the right notion for capacity-balanced descent.

D.3 Closure under the canonical core

Proposition D.1 (Fractal instantiation). *With $S = S_{\text{frac}}$, $G = G_{\text{frac}}$, and locally Lipschitz F_{base} , assumptions (A1)–(A4) hold on the open set $\Omega := \mathbb{R}^n \setminus (\mathcal{A}_{\delta_L} \cup \text{Cut}(\mathcal{A}))$, where $\text{Cut}(\mathcal{A})$ is the medial axis of $\mathcal{A}$. On Ω each d_{δ_ℓ} is C^2 and Theorems 9.1–9.4 apply with the rate constant of Theorem 9.3. Globally, $d_{\delta_\ell} \in C^{1,1}$, and (A1) can be replaced by (A1') ($S \in C^{1,1}$ with locally bounded weak Hessian) while preserving Theorems 9.1 and 9.2.*

Proof. On Ω the squared-distance functions are C^2 by Federer’s tubular-neighbourhood theorem; G_{frac} is constant SPD; apply Theorem 3.2. The $C^{1,1}$ extension follows from Rademacher and standard a.e. chain-rule arguments. $\square$

Proposition D.2 (Multi-scale exponential decay). *For $F_{\text{base}} \equiv 0$ on Ω ,*

$$E_{\text{frac}}(X(t)) \leq E_{\text{frac}}(X_0) e^{-\rho_{\text{frac}} t}, \quad \rho_{\text{frac}} = 4\sigma_J^2 \frac{\delta_L^{-4d_H}}{\delta_1^{-2d_H}} (1 - \gamma_{\max}),$$

with σ_J the smallest singular value of the Jacobian of S_{frac} on the operating sublevel set.

Proof. Apply Theorem 9.3 with $\mu_G = \delta_L^{-2d_H}$ and $M_G = \delta_1^{-2d_H}$ (eigenvalues of the diagonal G_{frac}). $\square$

E Contact-Geometric Extension

E.1 Motivation

Many controlled systems live on a contact manifold (M^{2m+1}, α) with $\alpha \wedge (d\alpha)^m \neq 0$. The Reeb field R_α and the contact distribution $\xi = \ker \alpha$ encode admissible motions. We show that the canonical ODE respects α when F_{base} is Reeb-compatible, without modifying the dynamics.

E.2 Contact instantiation

Work in $\mathbb{R}^{2m+1}$ with the standard contact form $\alpha = dz - \sum_{i=1}^m y_i dx_i$, coordinates (x, y, z) . (Throughout Part II, the symbol α denotes this contact 1-form; the projection coefficient of §B.2 is denoted $\alpha_{\text{proj}} = \langle F, g_X \rangle / \|g_X\|^2$ when both appear together.) Then $R_\alpha = \partial_z$ and $\xi = \text{span}\{\partial_{x_i} + y_i \partial_z, \partial_{y_i}\}$.

C.1 — Contact feature map

Let $\Phi : \mathbb{R}^{2m+1} \rightarrow \mathbb{R}^{k_0}$ encode the target submanifold (e.g. a Legendrian) as $\Phi = 0$. Define

$$S_{\text{ct}}(X) := \begin{pmatrix} \Phi(X) \\ \alpha_X(F_{\text{base}}(X)) \end{pmatrix} \in \mathbb{R}^{k_0+1},$$

where $\alpha_X(v) = v_z - \sum y_i v_{x_i}$. The metric is $G_{\text{ct}} = \text{blockdiag}(G_0, \beta)$, $\beta > 0$.

The second component penalises F_{base} -energy along the Reeb direction (non-horizontal motion).

E.3 Horizontal descent property

Proposition E.1 (Contact preservation). *Let $\phi(X) := \alpha_X(F_{\text{base}}(X)) = (F_{\text{base}})_z - \sum_i y_i (F_{\text{base}})_{x_i}$ and $L_\phi := \sup_X \|\nabla_X \phi(X)\| < \infty$ on the operating region. With $S = S_{\text{ct}}$ and $G = G_{\text{ct}}$ in the canonical ODE,*

$$\frac{d}{dt}(\beta \phi(X)^2) = 2\beta \phi(X) \langle \nabla_X \phi(X), \dot{X} \rangle \leq 2\beta L_\phi |\phi(X)| \|\dot{X}\|.$$

Moreover, by Lemma 7.7 applied to E_{ct} , the total energy obeys $\dot{E} \leq -(1 - \gamma)\|g_X\|^2 + (1 - \gamma)\langle g_X, F_{\text{base}} \rangle$, which (since the second component of g_X involves $\nabla_X \phi$) forces $\beta \phi(X)^2 \rightarrow 0$ whenever the first-component error $\Phi(X) \rightarrow 0$ and F_{base} is autonomous with ϕ vanishing at the target.

Proof. The identity is immediate from the chain rule $\dot{\phi} = \langle \nabla_X \phi, \dot{X} \rangle$ and Cauchy–Schwarz. The asymptotic statement follows by combining the energy descent of Theorem 9.1 (which forces $g_X \rightarrow 0$ along the trajectory in the autonomous case) with the structural fact that the Reeb-component of g_X is $2\beta \phi(X) \nabla_X \phi(X)$, so $g_X \rightarrow 0$ and a non-degeneracy of $\nabla_X \phi$ at the target force $\phi \rightarrow 0$. $\square$

Reading. The contact extension uses the canonical ODE *as-is*; preservation is purely a property of S_{ct} . Rule 2 (Euclidean projection) is preserved.

E.4 Subriemannian special case

If F_{base} is horizontal ($\alpha(F_{\text{base}}) = 0$), the second component of E_{ct} vanishes identically and the system reduces to the descent of Part I restricted to ξ . This recovers *subriemannian* gradient flow without modifying the canonical ODE.

F Geometric-Flow Extension

F.1 Time-varying metric

The frozen core requires G constant. We now allow $G = G(t)$ to evolve by a prescribed *geometric flow* (Ricci-type, mean-curvature, Yamabe), driven by the current state.

GF.1 — Metric flow

Let $\text{Ric}(G; X) \in \mathbb{R}_{\text{sym}}^{k \times k}$ be a smooth functional. The metric evolves by

$$\dot{G} = -2\text{Ric}(G; X) + \sigma (G_\star(X) - G), \quad \sigma \geq 0,$$

with $G_\star(X)$ a target metric (e.g. data-adaptive covariance). $\sigma = 0$ recovers pure Ricci flow; $\sigma > 0$ adds attractive damping toward $G_\star$.

F.2 Modified energy derivative

Lemma F.1 (Energy derivative under metric flow). *With $G = G(t)$ as above and $E(X, t) = S(X)^T G(t) S(X)$,*

$$\dot{E} = (1 - \gamma)(\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2) + S^T \dot{G} S.$$

Proof. $\dot{E} = 2S^T G \dot{S} + S^T \dot{G} S = \langle g_X, \dot{X} \rangle + S^T \dot{G} S$; apply Lemma 7.7 for the first piece. $\square$

For pure Ricci flow $S^T \dot{G} S = -2S^T \text{Ric} S$, which is ≤ 0 whenever $\text{Ric} \succeq 0$.

F.3 Joint stability

Theorem F.2 (Coupled state-metric stability). *Assume (A1),(A3),(A4), the relaxed metric assumption (A2'), $F_{\text{base}} \equiv 0$, $\sigma > 0$, and that $G_\star \in C^1$ with eigenvalues uniformly in $[\mu_\star, M_\star]$ and $\|\nabla_X G_\star\|_{\text{op}} \leq L_\star$. Assume $\text{Ric}(G; X) \succeq -c_R G$ with $c_R < \sigma/2$. If $G_\star$ is constant, the coupled system (X, G) is forward-complete and the Lyapunov function*

$$V(X, G) = E(X, t) + \frac{1}{2}\sigma^{-1}\|G - G_\star(X)\|_F^2$$

is non-increasing. If $G_\star$ depends on X , the same calculation gives the practical bound

$$\dot{V} \leq -a\|G - G_\star(X)\|_F^2 + \frac{L_\star^2}{4a\sigma^2}\|\dot{X}\|^2$$

for every $a \in (0, 1)$ after absorbing the Ricci term by $c_R < \sigma/2$.

Sketch. Lemma F.1 gives $\dot{E} \leq -(1 - \gamma)\|g_X\|^2 + 2c_R S^T G S + \sigma S^T (G_\star - G) S$. Differentiating the metric piece and using $\dot{G}_\star = (\nabla_X G_\star) \dot{X}$,

$$\frac{d}{dt} \frac{1}{2} \sigma^{-1} \|G - G_\star\|_F^2 = -\|G - G_\star\|_F^2 - 2\sigma^{-1} \langle G - G_\star, \text{Ric} \rangle_F - \sigma^{-1} \langle G - G_\star, \dot{G}_\star \rangle_F.$$

The last term is bounded by $\sigma^{-1} \|G - G_\star\|_F L_\star \|\dot{X}\|$. If $G_\star$ is constant this term vanishes, and adding to $\dot{E}$ while using $c_R < \sigma/2$ yields $\dot{V} \leq 0$. If $G_\star$ varies with X , Young's inequality gives the stated practical bound. $\square$

F.4 Discretisation

A practical scheme alternates a state half-step (canonical ODE with frozen G_t) and a metric half-step (one explicit Euler step of the Ricci flow). Convergence in the limit $h \rightarrow 0$ to the continuous coupled system follows from Lie–Trotter-type splitting estimates.

G Unified View of the Three Extensions

The consolidated table for Parts I–III is given in Section M at the end of Part III. The three v3 extensions all preserve the locked canonical ODE: Fractal and Contact change only S ; Geometric flow lifts (A2) to (A2').

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In every v3 extension $g_X = 2J^T GS$, the projection is Euclidean, and Rules 1–4 of Part I are obeyed.

Open extension targets (v3 candidates promoted to Part III)

The v3 candidate *information-geometric flow* is realised in Part III. The remaining v5 candidate list is consolidated at the end of Part III.

Part 2. Information-Geometric Flow**LOCKED**

v4 principle. Specialise the v3 metric flow with the *Fisher information metric*. Two coupling schedules are used: the *tracking* schedule $G \equiv \mathcal{I}_S(S(X))$ for structural results, and the *relaxation* schedule $\dot{G} = \sigma_{\text{rel}}(\mathcal{I}_S(S(X)) - G)$ for Lyapunov / practical state-metric control (specialisation of v3's Theorem F.2). The locked canonical ODE, the gradient formula $g_X = 2J^T GS$, and the Euclidean projection are unchanged. New *auxiliary* object: the state natural gradient $\widehat{g}_X = \mathcal{I}_X^\dagger g_X$, where $\mathcal{I}_X = J^T \mathcal{I}_S J$, which drives a chart-covariant variant flow (extension, not part of the locked system). Capabilities added: adaptive preconditioning, statistical Cramér–Rao rate floor, MLE/VI as canonical instances.

H Statistical setup and the Fisher metric**H.1 Features as statistical parameters**

Identify the feature coordinate $z = S(X) \in \mathbb{R}^k$ with the parameter of a smooth parametric family of probability densities

$$\mathcal{P} = \{ p(\cdot | z) : z \in \Theta_S \subset \mathbb{R}^k \}, \quad p(\cdot | z) > 0, \quad \int p dy = 1.$$

The feature score is $u(y; z) := \nabla_z \log p(y | z)$.

IG.1 — Fisher information matrix

$$\mathcal{I}_S(z) := \mathbb{E}_{y \sim p(\cdot | z)} [u(y; z) u(y; z)^T] = -\mathbb{E}[\nabla_z^2 \log p(y | z)] \in \mathbb{R}^{k \times k}.$$

$\mathcal{I}_S(z) \succeq 0$ always; positive-definite under the standard identifiability assumption (**A5**: feature scores span $\mathbb{R}^k$ at every $z = S(X)$ in the operating region). The pullback Fisher metric on state space is

$$\mathcal{I}_X(X) := J(X)^T \mathcal{I}_S(S(X)) J(X) \in \mathbb{R}^{n \times n}.$$

We henceforth assume **A5**; when $\mathcal{I}_X$ is singular, all inverses below are Moore–Penrose inverses or Tikhonov-regularised inverses on $\text{range}(J^T)$.

H.2 Fisher metric in the canonical system

Under (A2') we may take $G(X) := \mathcal{I}_S(S(X))$. The canonical objects then read

$$E(X) = S(X)^T \mathcal{I}_S(S(X)) S(X), \quad g_X = 2J(X)^T \mathcal{I}_S(S(X)) S(X),$$

and the canonical ODE, the Euclidean projection (Rule 2), and Rules 1, 3, 4 are unchanged. We adopt either of two coupling schemes, used below as needed:

- (F1) **Tracking schedule.** $G(X) \equiv \mathcal{I}_S(S(X))$ (instantaneous identification). Then $\dot{G}$ is the chain-rule derivative $\nabla_X \mathcal{I}_S(S(X))[\dot{X}]$ and the v3 flow equation does not apply.

- (F2) **Relaxation schedule.** $G(t)$ evolves independently by $\dot{G} = \sigma_{\text{rel}}(\mathcal{I}_S(S(X)) - G)$ with relaxation rate $\sigma_{\text{rel}} > 0$. The v3 joint-stability theorem applies with $G_\star(X) = \mathcal{I}_S(S(X))$, $\text{Ric} \equiv 0$, subject to the state-dependent-target qualifications in Theorem F.2. In the limit $\sigma_{\text{rel}} \rightarrow \infty$ this recovers (F1).

Throughout Part III we use (F1) for structural results and (F2) for the joint-stability statement (Theorem J.1).

I Natural gradient and chart covariance

NG.1 — Natural gradient

With $G(X) = \mathcal{I}_S(S(X))$, define the state natural gradient of E as

$$\widetilde{g}_X(X) := \mathcal{I}_X(X)^\dagger g_X(X), \quad \mathcal{I}_X = J^T \mathcal{I}_S J.$$

$\widetilde{g}_X$ is an auxiliary quantity used *only* for analysis and for the *natural-gradient variant* of the canonical ODE introduced in Section K. The locked canonical ODE itself is unchanged and continues to use g_X .

Theorem I.1 (Chart covariance of $\widetilde{g}_X$). *Under a C^1 diffeomorphism $X = \Phi(\tilde{X})$ with Jacobian $T = \partial\Phi/\partial\tilde{X}$, the state Fisher metric and the natural gradient transform as*

$$\tilde{\mathcal{I}}_X = T^T \mathcal{I}_X T, \quad \tilde{\widetilde{g}}_{\tilde{X}} = T^{-1} \widetilde{g}_X,$$

so the natural-gradient flow $\dot{X} = -\widetilde{g}_X(X)$ is covariant: its trajectory in $\tilde{X}$ -coordinates is $\Phi^{-1} \circ X(\cdot)$. Equivalently, the trajectory is chart-independent.

Proof. $\tilde{\mathcal{I}}_X = T^T \mathcal{I}_X T$ is the standard transformation (Chentsov–Cencov for the Fisher metric; [8]). Then $\tilde{\nabla} \tilde{E} = T^T \nabla E$, so $\tilde{\widetilde{g}} = \tilde{\mathcal{I}}_X^\dagger \tilde{\nabla} \tilde{E} = (T^T \mathcal{I}_X T)^\dagger T^T \nabla E = T^{-1} \mathcal{I}_X^\dagger \nabla E = T^{-1} \widetilde{g}_X$. Covariance of the flow follows. $\square$

Remark I.2 (The locked canonical ODE is *not* chart-covariant). The canonical ODE of Part I uses the Euclidean inner product (Rule 2) in the projection term. Under reparametrisation this inner product is not preserved, so the canonical trajectory $X(t)$ is chart-dependent. Chart covariance is a property of the *auxiliary* natural-gradient flow $\dot{X} = -\widetilde{g}_X$ and of stability statements formulated intrinsically in the Fisher metric (see Theorem J.1); it is not a property of the locked ODE.

Computational consequence. The natural-gradient flow $\dot{X} = -\widetilde{g}_X$ is the invariant analogue of canonical descent. Even though the locked canonical ODE remains chart-dependent, working in (locally) Fisher-affine coordinates makes Euclidean projection nearly optimal and removes the conditioning problems that plague v2/v3.

J Joint stability under Fisher flow

Theorem J.1 (Information-geometric stability). *Assume (A1),(A2'),(A3)–(A5), $F_{\text{base}} \equiv 0$, the relaxation schedule (F2) with $\sigma_{\text{rel}} > 0$, and that $\mathcal{I}_S(S(X))$ is uniformly bounded $\mu_{\mathcal{I}} I \preceq \mathcal{I}_S(S(X)) \preceq M_{\mathcal{I}} I$ on the operating sublevel set, with $\nabla_X \mathcal{I}_S(S(X))$ uniformly bounded in operator norm by $L_{\mathcal{I}}$. If $\mathcal{I}_S$ is constant on the operating set, then the Lyapunov function*

$$V(X, G) = E(X) + \frac{1}{2} \sigma_{\text{rel}}^{-1} \|G - \mathcal{I}_S\|_F^2$$

is non-increasing, and along trajectories

$$E(X(t)) \leq E(X_0) e^{-\rho_F t}, \quad \rho_F \geq 4 \frac{\mu_{\mathcal{I}}^2}{M_{\mathcal{I}}} \sigma_{\text{min}}^2(J) (1 - \gamma_{\text{max}}).$$

The displayed Euclidean rate is chart-dependent. By Theorem I.1, however, the auxiliary natural-gradient flow is chart-covariant, so stability statements formulated intrinsically with the pullback Fisher metric $\mathcal{I}_X$ are chart-independent. If $\mathcal{I}_S$ varies with X , the same V satisfies the practical bound of Theorem F.2 with $G_\star(X) = \mathcal{I}_S(S(X))$ rather than strict monotonicity.

Proof. The Lyapunov inequality is Theorem F.2 with $G_\star = \mathcal{I}_S$, $\text{Ric} \equiv 0$ in the constant case. If $\mathcal{I}_S$ varies, Theorem F.2 supplies the displayed practical bound. The decay rate follows from the corrected Theorem 9.3 with $\mu_G = \mu_{\mathcal{I}}$, $M_G = M_{\mathcal{I}}$. Chart covariance of the auxiliary natural-gradient flow is Theorem I.1. $\square$

Remark J.2 (Cramér–Rao floor). If the family $\mathcal{P}$ admits an unbiased estimator with covariance bounded above by $\text{Cov}_{\max} I$, the Cramér–Rao inequality gives $\mu_{\mathcal{I}} \geq \text{Cov}_{\max}^{-1}$, so $\rho_F \geq 4 \text{Cov}_{\max}^{-2} M_{\mathcal{I}}^{-1} \sigma_{\min}^2(J)(1 - \gamma_{\max})$. This is a *statistical* lower bound on the descent rate that v2/v3 cannot express.

K Three new computational capabilities

K.1 Adaptive preconditioning and natural-gradient variant

The feature Fisher matrix $\mathcal{I}_S(S(X))$ and its pullback $\mathcal{I}_X = J^T \mathcal{I}_S J$ are the local information preconditioners for optimisation on $\mathcal{P}$. Two ways to use them:

- (1) **In-place (locked ODE).** Set $G(X) = \mathcal{I}_S(S(X))$. The locked ODE runs unchanged, with $g_X = 2J^T \mathcal{I}_S S$. This preserves Rules 1–4 but is chart-dependent.
- (2) **Natural-gradient variant.** Replace the locked ODE by $\dot{X} = -\widetilde{g}_X$ (Definition NG.1). Chart-covariant (Theorem I.1), but *breaks Rule 2* and is therefore an extension, not a specialisation. Use only when chart invariance is required.

K.2 Variational inference and MLE in one map

Two canonical instances:

- (1) **MLE.** In local asymptotic-normal coordinates, choose $S(X) := X - X_{\text{MLE}}$ and $G = \mathcal{I}_S$. Then $E = S^T \mathcal{I}_S S$ is the quadratic log-likelihood error; the auxiliary natural-gradient variant is locally Fisher scoring.
- (2) **Variational inference.** Choose $S(X) := \lambda(X) - \lambda^\star$ in natural-parameter coordinates of q_X . With $G = \mathcal{I}_S$, E is the local Fisher quadratic approximation to the KL/negative ELBO; the auxiliary flow $\dot{X} = -\widetilde{g}_X$ is natural-gradient VI.

K.3 Information-geometric Hessian

Under (A2') with $G = \mathcal{I}_S(S(X))$, the flat Euclidean Hessian of $E = S^T \mathcal{I}_S(S(X))S$ acquires extra terms beyond the constant-metric formula of §C.5:

$$(\nabla_X^2 E)_{ab} = 2(J^T \mathcal{I}_S J)_{ab} + \eta_i K^i_{ab} + 2(\partial_a \mathcal{I}_{S,ij}) J^j_b S^i + 2(\partial_b \mathcal{I}_{S,ij}) J^j_a S^i + S^i (\partial_a \partial_b \mathcal{I}_{S,ij}) S^j,$$

where $\eta = 2\mathcal{I}_S S$. The two single-derivative terms involve the Fisher-information *connection coefficients* $C_{abc} := \frac{1}{2} \partial_a \mathcal{I}_{S,bc}$ (Amari 0-connection); the last term involves $\partial^2 \mathcal{I}$, which vanishes for exponential families in canonical coordinates. In particular, on the target set $\{S = 0\}$ (where $\eta = 0$ and the last three corrections vanish), $\nabla_X^2 E = 2J^T \mathcal{I}_S J = 2\mathcal{I}_X$ recovers the Fisher pulled-back metric at the optimum.

L Bridge to v5: stochastic extension

For a canonical SDE $dX = H(X) dt + \Sigma(X) dW$ (with H the canonical-ODE drift), Itô's lemma gives

$$\mathbb{E}[dE] = \langle g_X, H \rangle dt + \frac{1}{2} \text{tr}(\Sigma^T \nabla_X^2 E \Sigma) dt.$$

The Fisher choice $\Sigma = \mathcal{I}_X^{\dagger/2}$ makes the second term $\frac{1}{2}\text{tr}((\mathcal{I}_X^{\dagger/2})^T \nabla_X^2 E \mathcal{I}_X^{\dagger/2})$, which on the target set ($\eta = 0$ and $\nabla_X^2 E = 2\mathcal{I}_X$) reduces to $\text{rank}(\mathcal{I}_X)$, a chart-independent scalar on the identifiable tangent subspace. This prepares v5 by:

- providing the canonical noise covariance $\Sigma = \mathcal{I}_X^{\dagger/2}$,
- yielding chart-independent Itô corrections,
- identifying V with a free-energy functional.

M Master unified view (v2 + v3 + v4)

Layer	Object changed	Canonical ODE	New result / capability
Part I (v2)	frozen kernel (S, G const)	defined	descent, exp. rate, ult. bound- edness
Part II (v3)	S via $S_{\text{frac}}, S_{\text{ct}}$	unchanged	fractal / contact
Part II (v3)	$G \rightarrow G(t)$ Ricci-type flow	unchanged	Lyapunov / practical state- metric bound (Thm. F.2)
Part III (v4)	$G(X) = \mathcal{I}_S(S(X)) + \text{relaxation}$	unchanged (locked)	adaptive pre- conditioning, Cramér–Rao floor
Part III (v4)	auxiliary natural-gradient flow	<i>new variant</i> (extension)	chart covariance (Thm. I.1)

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In every *locked-ODE* row $g_X = 2J^T GS$, the projection is Euclidean, and Rules 1–4 of Part I are obeyed. The natural-gradient variant in the last row *breaks Rule 2* and is explicitly marked as an extension.

Open extension targets (v5 candidates)

- Stochastic canonical SDE with Fisher-noise $\Sigma = \mathcal{I}_X^{\dagger/2}$.
- Symplectic / Poisson extension via skew-symmetric energy correction.
- Discrete fractal lattices: S defined on a self-similar graph.
- α -connection family: interpolate between Fisher and dual-flat geometry.

N Thermodynamic dictionary

The damping law $\gamma = E/(E + \theta)$ is structurally a Fermi–Dirac occupation in the dimensionless ratio E/θ :

$$\gamma = \frac{1}{1 + \theta/E}, \quad 1 - \gamma = \frac{1}{1 + E/\theta}.$$

Hence θ plays the role of a *temperature* (units of energy) and the canonical loop has an exact thermodynamic reading.

Canonical thermodynamic quantities

Temperature:	θ .
Occupation:	$\gamma = E/(E + \theta)$.
Free energy:	$F(E) = E - \theta \mathcal{S}(E)$.
Entropy:	$\mathcal{S}(E) = -\gamma \ln \gamma - (1 - \gamma) \ln(1 - \gamma)$.
Carnot-like efficiency:	$\eta_C = 1 - \gamma$.
Heat dissipated:	$Q = \gamma \text{proj}_{g_X} F_{\text{base}}$.
Work preserved:	$W = F_{\text{base}} - Q$.

Proposition N.1 (Maximum-entropy operating point). $\mathcal{S}(E)$ is strictly concave on $(0, \infty)$ with maximum at $E = \theta$, where $\gamma = 1/2$ and $\mathcal{S} = \ln 2$.

Proof. $\mathcal{S}$ is the binary entropy of $\gamma \in (0, 1)$; $\gamma = 1/2 \iff E = \theta$. $\square$

Remark N.2 (Annealing). Decreasing θ over time is *simulated annealing* on the canonical surface: $\gamma \rightarrow 1$, the engine commits more strongly to projection, and ρ_{eff} saturates Theorem 9.3. Increasing θ relaxes the brake and exposes more of F_{base} .

Proposition N.3 (de Bruijn-type identity, Fisher case). Under the Fisher choice $G = \mathcal{I}_S$ and the regularity of Section H,

$$\dot{E}(t) = -\rho_{\text{eff}}(t) E(t), \quad \rho_{\text{eff}}(t) \leq \rho_{\text{CR}},$$

so the engine trades Lyapunov energy for Fisher information at a rate bounded by the Cramér–Rao floor.

O Game-theoretic derivation of γ^*

The damping coefficient is the Nash equilibrium of a two-player saddle-point problem between an *actor* (executing F_{base}) and an *auditor* (enforcing $S \rightarrow 0$).

Theorem O.1 (Saddle-point characterisation). Fix F_{base} and g_X with $\|g_X\| > 0$. Consider

$$\mathcal{L}(u, \lambda) = \frac{1}{2} \|u - F_{\text{base}}\|^2 + \lambda \langle u, g_X \rangle - \frac{\theta}{2} \frac{\|g_X\|^2}{E} \lambda^2.$$

Then $\mathcal{L}$ has a unique saddle point (u^*, λ^*) with

$$u^* = F_{\text{base}} - \gamma \frac{\langle F_{\text{base}}, g_X \rangle}{\|g_X\|^2} g_X, \quad \gamma = \frac{E}{E + \theta},$$

which coincides exactly with the canonical step.

Proof. Stationarity in u gives $u^* = F_{\text{base}} - \lambda^* g_X$. Stationarity in λ gives $\langle u^*, g_X \rangle = (\theta \|g_X\|^2 / E) \lambda^*$, and substituting yields $\lambda^* = \langle F_{\text{base}}, g_X \rangle E / ((E + \theta) \|g_X\|^2)$, so u^* matches the canonical step with $\gamma = E / (E + \theta)$. The Hessian of $\mathcal{L}$ in u is $I \succ 0$ and in λ is $-\theta \|g_X\|^2 / E < 0$, so the stationary point is a saddle. $\square$

Remark O.2 (No-regret reading). Because u^* is a saddle of $\mathcal{L}$, the canonical step is *minimax-optimal*: among admissible perturbations of F_{base} , none yields a worst-case energy increase greater than zero. The canonical engine is therefore an adversarial robustifier: any input field F_{base} is converted into a no-regret update against the energy-projection auditor.

P Multi-engine composition (port-Hamiltonian)

Multiple canonical engines, each defined by its own triple $(S_i, G_i, F_{\text{base}}^{(i)})$ with energy $E_i = S_i^T G_i S_i$ and damping $\gamma_i = E_i/(E_i + \theta_i)$, can be aggregated into a single passive port-Hamiltonian system on the common state $X \in \mathbb{R}^n$.

Definition P.1 (Composite engine). For each $i = 1, \dots, m$ let $R_i(X) = \gamma_i(X) g_X^{(i)} (g_X^{(i)})^T / \|g_X^{(i)}\|^2 \succeq 0$ and $H(X) = \frac{1}{2} \sum_i E_i(X)$. Define

$$\dot{X} = \sum_{i=1}^m F_{\text{base}}^{(i)}(X) - 2 \left(\sum_{i=1}^m R_i(X) \right) \nabla H(X).$$

Theorem P.2 (Aggregate passivity). Under Assumptions A1–A3 applied component-wise, the composite engine of Definition P.1 satisfies

$$\dot{H} \leq \sum_{i=1}^m \langle \nabla H, F_{\text{base}}^{(i)} \rangle,$$

i.e. it is passive with input $u = \sum_i F_{\text{base}}^{(i)}$ and storage H . In particular, if every $F_{\text{base}}^{(i)} \equiv 0$ then $\dot{H} \leq 0$ and each E_i is non-increasing along the joint flow.

Proof. $\dot{H} = \sum_i \dot{E}_i/2$. From $\dot{E}_i = \langle g_X^{(i)}, \dot{X} \rangle$ and the substitution of the composite ODE, $\dot{E}_i = \langle g_X^{(i)}, \sum_j F_{\text{base}}^{(j)} \rangle - 2 \sum_j \langle g_X^{(i)}, R_j \nabla H \rangle$. Each $R_j \succeq 0$ contributes a non-positive term to $\sum_i \dot{E}_i$, hence $\dot{H} \leq \langle \nabla H, \sum_j F_{\text{base}}^{(j)} \rangle$. $\square$

Remark P.3 (Strategy aggregation). Theorem P.2 states that running several canonical engines in parallel on the same state is automatically stability-preserving: no portfolio-level Lyapunov function is needed, and individual energies E_i remain non-increasing in the free-flow regime. This is the canonical justification for combining multiple (S, G, F_{base}) instantiations (e.g. several trading strategies, or coupled control objectives) without redesign.

Remark P.4 (Single-temperature limit). If all $\theta_i \equiv \theta$ and the $g_X^{(i)}$ are mutually orthogonal, then $R = \sum_i R_i$ has rank m and the composite engine acts as m independent canonical engines sharing one temperature; the total entropy is $\mathcal{S}_{\text{tot}} = \sum_i \mathcal{S}(E_i)$.

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